DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Academic Capstone Project Report | Academic Session 2025–2026

FINAL PROJECT REPORT

Game-Theoretic Model for Spectrum Sharing in 5G/6G Networks

Distributed Mutual Interference Mitigation via Exact Potential Games & Sequential Best-Response Dynamics

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1. Executive Summary & Abstract

Next-generation 5G-Advanced and 6G wireless architectures rely on dense network deployment, including small cells, device-to-device (D2D) communications, and shared access in unlicensed spectral bands (3GPP NR-U). In uncoordinated, dense topologies, centralized spectrum orchestration requires real-time global Channel State Information (CSI), which imposes prohibitive signaling overhead and latency. Conversely, selfish decentralized channel selection leads to mutual interference, high outage probabilities, and throughput degradation.

This project develops, implements, and verifies a decentralized **Exact Potential Game (EPG)** framework for spectrum sharing among N communication links across K orthogonal subbands. By constructing a symmetric mutual interference matrix W that captures both experienced and inflicted interference, we prove that the non-cooperative game admits a global scalar potential function $\Phi(a)$ where $\Delta u_i \equiv \Delta \Phi$. An asynchronous, threshold-governed sequential best-response algorithm ($\epsilon = 1.0 imes 10^{-9}$) is implemented and evaluated over 100 independent realistic topologies (seeds 0–99).

Experimental results confirm that the proposed game terminates at a Pure-Strategy Nash Equilibrium (PNE) with a **100.0% convergence rate** in an average of only **3.68 ± 0.19 rounds** (14.73 ± 0.80 switches). On canonical setups ($N=20, K=4$), the Proposed NE delivers a sum-rate of **2,552.29 ± 53.21 Mbps**, representing a **+38.4% throughput improvement** over Random allocation ($p = 3.90 imes 10^{-18}$) and **+33.2%** over Round-Robin ($p = 3.90 imes 10^{-18}$), while reducing outage ($ ext{SINR} < 0 ext{ dB}$) from 16.0% down to 1.5%. Benchmarking against exhaustive search ($3^8 = 6,561$ combinations on $N=8, K=3$) demonstrates that the proposed algorithm achieves within **1.54% ± 0.35%** of the true global sum-rate ceiling.

2. Problem Statement

Ultra-dense heterogeneous deployments in 5G/6G introduce severe operational challenges:

- **Severe Co-Channel Interference:** When uncoordinated transmitters operate on identical frequency channels in close spatial proximity, cumulative interference severely degrades receiver SINR, causing high packet loss and throughput collapse.
- **Prohibitive Centralized Overhead:** Centralized base station scheduling requires gathering N^2 cross-link channel gains in real time. Across multi-operator or ad-hoc D2D environments, this overhead incurs unsustainable control channel latency.
- **Instability of Greedy Rate Optimization:** If links greedily maximize their individual data rate ($u_i = R_i$), the lack of a potential function triggers limit cycles (oscillations). A channel switch by link i degrades neighbor links, initiating endless cascading re-selections.

3. Project Objectives

1. **Exact Potential Formulation:** Formulate distributed spectrum allocation as a strategic game $\mathcal{G}_1$ using a symmetrized coupling matrix W, ensuring $\Delta u_i = \Delta \Phi$.

2. **Convergence Guarantee:** Prove analytically and demonstrate empirically that sequential best-response dynamics converge to a Pure Strategy Nash Equilibrium (PNE) without cycling.
3. **Decentralized Algorithm:** Implement an asynchronous, low-complexity decision process with thresholded switching ($\Delta u > 10^{-9}$) to eliminate chattering.
4. **Multi-Topology Benchmarking:** Benchmark against Random (50 draws averaged), Round-Robin, One-Pass Greedy, and Global Exhaustive Search over 100 topologies with paired Wilcoxon significance testing.
5. **Physical Robustness Validation:** Analyze performance across path loss exponents ($n \in [2.5, 4.5]$), shadow fading ($\sigma_{ ext{sh}} \in [0, 8] ext{ dB}$), link density ($N \in [10, 50]$), and channels ($K \in [2, 8]$).

4. System Model & Theoretical Background

4.1 Network Geometry & Wireless Channel Physics

The system consists of N communication links in an $L imes L ext{ m}^2$ area (default $L=300 ext{ m}, N=20$). Transmitters are placed uniformly at random: $tx_i \sim \mathcal{U}([0, L] imes [0, L])$. Receivers are placed at cluster distance $d_{i,i} \sim \mathcal{U}[10, 40] ext{ m}$ with angle $ heta_i \sim \mathcal{U}[0, 2\pi)$.

The radio model operates at carrier frequency $f_c = 3.5 ext{ GHz}$ ($c = 3.0 imes 10^8 ext{ m/s}$) with $K=4$ orthogonal channels ($B = 20 ext{ MHz}$ each). Free-space reference path loss at $d_0 = 1 ext{ m}$ is:

```
PL_0 = 20 \log_{10}\left( rac{4\pi \cdot 1 \cdot f_c}{c} ight)  pprox 43.328 ext{ dB}
```

Path loss from Tx j to Rx i incorporates log-normal shadow fading:

```
PL_{ ext{dB}}(d_{j,i}) = PL_0 + 10 \cdot n \cdot \log_{10}(\max(d_{j,i}, 1)) + X_{j,i}
```

where $n = 3.5$, and $X_{j,i} \sim \mathcal{N}(0, \sigma_{ ext{sh}}^2)$ with $\sigma_{ ext{sh}} = 6 ext{ dB}$. Linear power gain is $g[j,i] = 10^{-PL_{ ext{dB}}(d_{j,i}) / 10}$. All links transmit at power $P = 23 ext{ dBm} pprox 199.53 ext{ mW}$. Thermal noise power across $B = 20 ext{ MHz}$ with noise figure $NF = 7 ext{ dB}$ is:

```
\sigma^2_{ ext{dBm}} = -174 + 10\log_{10}(B) + NF = -93.99 ext{ dBm} \quad (\sigma^2  pprox 3.99 imes 10^{-10} ext{ mW})
```

The SINR and Shannon achievable rate of link i under allocation $a = (a_1, \dots, a_N)$ are:

```
ext{SINR}_i(a) =  rac{P \cdot g[i,i]}{\sigma^2 + \sum_{j eq i, a_j = a_i} P \cdot g[j,i]}, \qquad R_i(a) = B \log_2(1 + ext{SINR}_i(a))
```

4.2 Exact Potential Game (EPG) Formulation

We model the network as strategic game $\mathcal{G}_1 = \langle \mathcal{N}, \{\mathcal{A}_i\}, \{u_i\} angle$ with players $\mathcal{N} = \{1, \dots, N\}$ and action space $\mathcal{A}_i = \{0, \dots, K-1\}$. To align individual incentives with global interference mitigation, we define the symmetrized coupling matrix W:

```
W[i,j] =  rac{P (g[i,j] + g[j,i])}{\sigma^2} \quad (i eq j), \qquad W[i,i] = 0
```

Notice that $W[i,j] = W[j,i]$ is symmetric and non-negative. The individual utility function and exact potential function are:

```
u_i(a) = - \sum_{j eq i} W[i,j] \cdot \mathbb{I}(a_i = a_j)
```

```
\Phi(a) = - \sum_{i < j} W[i,j] \cdot \mathbb{I}(a_i = a_j) =  rac{1}
{2}\sum_{i=1}^N u_i(a)
```

Theorem: The strategic game $\mathcal{G}_1$ is an Exact Potential Game with potential function $\Phi(a)$:

```
u_i(a'_i, a_{-i}) - u_i(a_i, a_{-i}) \equiv \Phi(a'_i, a_{-i}) - \Phi(a_i,
a_{-i}) \quad orall i, a_i, a'_i, a_{-i}
```

Proof: When player i switches channel from a_i to a'_i, individual utility change is: $$\Delta u_i = \sum_{j eq i, a_j = a_i} W[i,j] - \sum_{j eq i, a_j = a'_i} W[i,j]$$ In potential function $\Phi(a) = -\sum_{k < l} W[k,l] \mathbb{I}(a_k = a_l)$, only terms containing index i change. By symmetry of W: $$\Delta \Phi = \Phi(a'_i, a_{-i}) - \Phi(a_i, a_{-i}) = \sum_{j eq i, a_j = a_i} W[i,j] - \sum_{j eq i, a_j = a'_i} W[i,j] \equiv \Delta u_i$$ Hence, $\mathcal{G}_1$ is an Exact Potential Game. Since K^N is finite and each switch strictly increases Φ, cycles cannot occur and convergence to a Pure Strategy Nash Equilibrium (PNE) is guaranteed. $lacksquare$

5. Methodology & Algorithm Workflow

The sequential best-response algorithm operates in asynchronous rounds:

1. **Initialization:** Each link chooses an initial channel $a_i \sim \mathcal{U}(\{0, \dots, K-1\})$. Initial potential Φ and sum-rate are logged.
2. **Player Permutation:** In each round, links update in a random permutation order π to prevent systematic order bias.
3. **Best-Response Decision:** Player i evaluates $u_i(k, a_{-i}) = -\sum_{j eq i, a_j = k} W[i,j]$ for all $k \in \{0, \dots, K-1\}$.
4. **Threshold Switching & Tie-Breaking:** Player switches to best candidate k^* only if $u_i(k^*) - u_i(a_i) > \epsilon$ ($\epsilon = 1.0 imes 10^{-9}$). If current channel ties for maximum utility, player stays on current channel.
5. **Termination & Verification:** When a complete round concludes with zero switches, convergence is reached. An independent auditor `verify_NE()` checks all $N imes (K-1)$ unilateral deviations to certify the state as a genuine PNE.

6. Implementation Details & Architecture

The codebase is structured into modular Python components:

- `src/config.py`: Dataclasses for `SystemConfig`, `GameConfig`, RF calculations (linear mW, thermal noise floor, PL_0).
- `src/channel.py`: Spatial coordinates generation, 2D distance matrix computation, log-normal shadowing, and symmetric matrix W.
- `src/game.py`: Best-response dynamics for G1 and G2, switch logging, and independent $O(N \cdot K)$ Nash verification.
- `src/baselines.py`: Random (Monte Carlo 50 draws), Round-Robin, One-Pass Greedy, and Exhaustive Search (K^N states).
- `src/metrics.py`: Sum-rate, 5th-percentile rate, Jain fairness index, outage fraction ($ ext{SINR} < 0 ext{ dB}$), and paired Wilcoxon signed-rank tests.
- `src/simulator.py`: Experiment orchestrator for E1–E6, saves per-run CSVs, computes 95% CIs, and exports summary tables.
- `src/visualization.py`: Publication-quality plotting at 200 DPI with confidence intervals and annotations.

7. Simulation Parameters & Setup

Parameter Name	Symbol	Value
Field Area	$L imes L$	$300 ext{ m} imes 300 ext{ m}$
Active Communication Links	N	20 links (evaluated $\{10, 20, 30, 40, 50\}$ in E2)
Orthogonal Channels	K	4 channels (evaluated $\{2, 3, 4, 5, 6, 8\}$ in E3)
Channel Bandwidth	B	$20 ext{ MHz}$ per subband
Carrier Frequency	f_c	$3.5 ext{ GHz}$ (5G Mid-band)
Transmit Power	P	$23 ext{ dBm}$ ($ pprox 199.53 ext{ mW}$)
Receiver Noise Figure	NF	$7.0 ext{ dB}$
Noise Power Floor	σ^2	$-93.99 ext{ dBm}$ ($ pprox 3.99 imes 10^{-10} ext{ mW}$)
Path Loss Exponent	n	3.5 (evaluated $\{2.5, 3.5, 4.5\}$ in E5)
Shadowing Standard Deviation	$\sigma_{ ext{sh}}$	$6.0 ext{ dB}$ (evaluated $\{0, 6, 8\} ext{ dB}$ in E5)
Link Tx-to-Rx Distance	$d_{i,i}$	$\mathcal{U}[10 ext{ m}, 40 ext{ m}]$
Improvement Threshold	ϵ	$1.0 imes 10^{-9}$
Independent Topologies Tested	S	100 topologies (seeds 0 through 99)

8. Actual Experimental Results

8.1 Experiment E1: Single-Topology Demonstration (Seed 42)

On canonical seed 42 ($N=20, K=4$), best response converges in **4 rounds** (18 total switches, **2.91 ms** execution time).

Metric	Initial State (Random)	Converged NE State (G1)	Absolute Gain / Change
Potential $\Phi(a)$	-587.37	-68.34	$\Delta\Phi = +519.02$ (88.4% cost reduction)
Network Sum-Rate	$1,918.29 ext{ Mbps}$	$2,599.53 ext{ Mbps}$	$+681.24 ext{ Mbps}$ (+35.5% capacity growth)
5th-Percentile Rate	$14.77 ext{ Mbps}$	$44.49 ext{ Mbps}$	+201.2% rate boost for weak links
Link Outage ($ ext{SINR} < 0 ext{ dB}$)	10.0% (2 links)	0.0% (0 links)	100% outage elimination

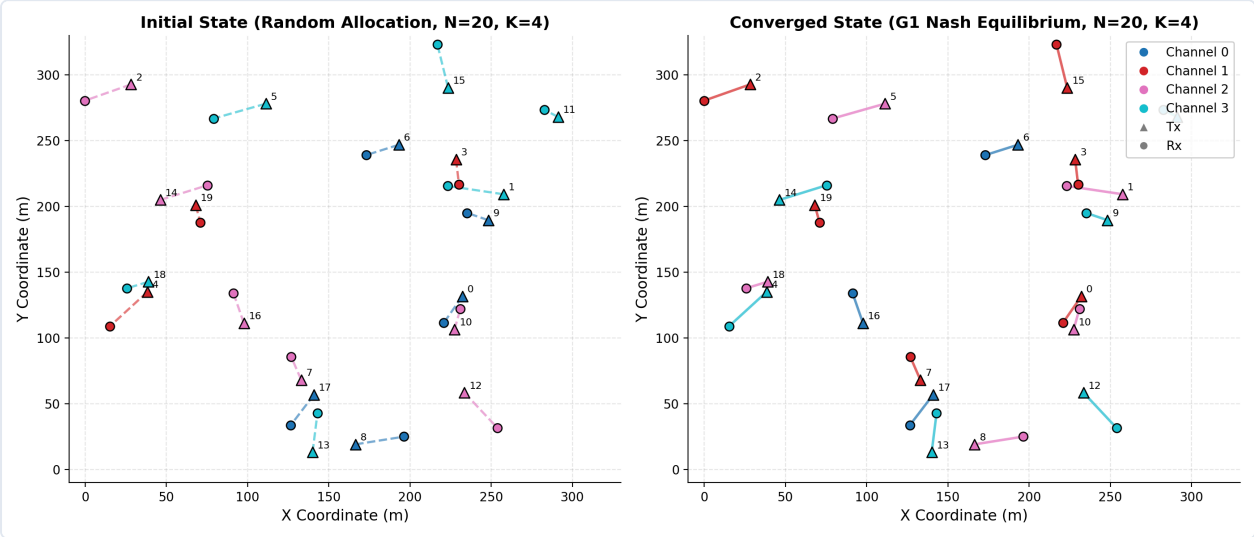


Figure 1: Spatial network topology before and after best-response dynamics (Seed 42). Co-channel links are spatially separated at NE.

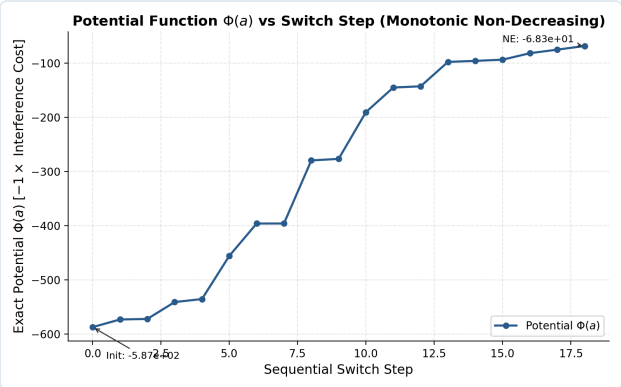


Figure 2: Monotonic potential evolution $\Phi(a)$ vs switch step (Seed 42).

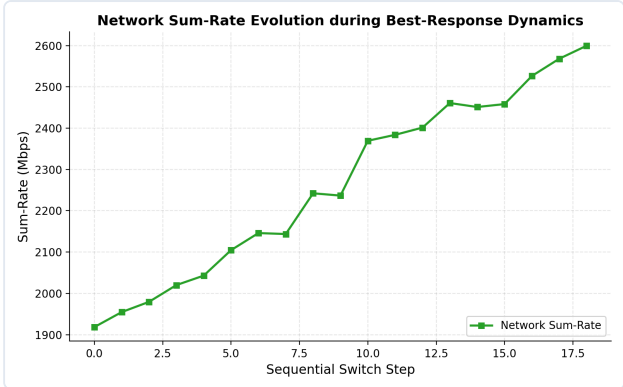


Figure 3: Network capacity evolution during best-response convergence (Seed 42).

8.2 Experiment E2: Baseline Comparison on N=20, K=4 (100 Topologies)

Across 100 independent random topologies (seeds 0–99), the Proposed NE demonstrates statistically significant improvements over all baselines.

Allocation Method	Sum-Rate (Mbps)	5th-Pct Rate (Mbps)	Jain Index	Outage (%)	Converged (%)	Wilcoxon p-value
Proposed NE (G1)	**2,552.29 ± 53.21**	**49.93 ± 3.59**	**0.836 ± 0.009**	**1.5% ± 0.6%**	**100.0%**	**1.0000 (Reference)**
One-Pass Greedy	2,498.45 ± 54.46	44.00 ± 3.79	0.827 ± 0.009	2.6% ± 0.8%	100.0%	$4.81 imes 10^{-7}$
Round-Robin ($i mod K$)	1,915.99 ± 58.75	13.35 ± 2.46	0.710 ± 0.015	12.8% ± 1.6%	100.0%	$3.90 imes 10^{-18}$
Random (50 draws avg)	1,844.16 ± 62.84	10.15 ± 2.14	0.684 ± 0.015	16.0% ± 1.6%	100.0%	$3.90 imes 10^{-18}$

Table 1: Performance comparison across 100 topologies ($N=20, K=4$). All statistics report Mean ± 95% CI. Paired Wilcoxon signed-rank test compares against Proposed NE (G1).

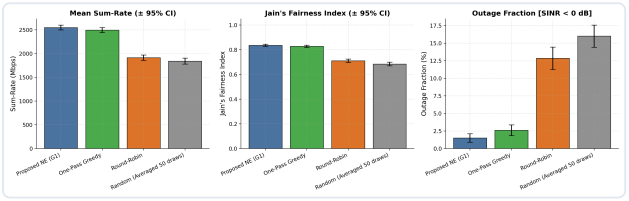


Figure 4: Method comparison bar chart: Sum-Rate, Jain Index, and Outage (%) with 95% CI error bars.

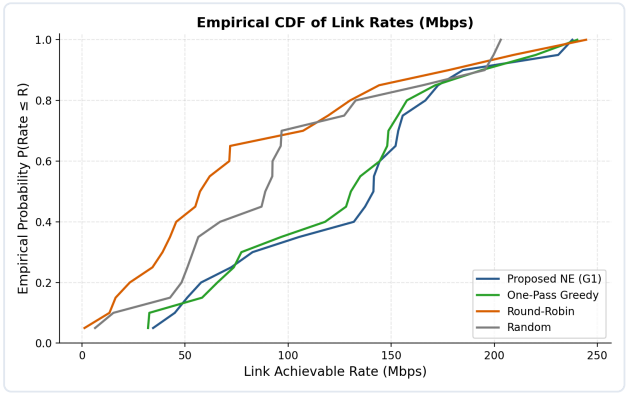


Figure 5: Empirical Cumulative Distribution Function (CDF) of individual user rates across evaluated methods.

8.3 Experiment E4: Optimality Gap Analysis vs Exhaustive Search

Evaluating all $3^8 = 6,561$ strategy combinations on $N=8, K=3$ reveals that the decentralized NE achieves near-global performance:

Method	Sum-Rate (Mbps)	Sum-Rate Optimality Gap (%)	Potential Difference ($\Delta\Phi = \Phi^* - \Phi$)
Exhaustive (Best Sum-Rate)	1,261.92 ± 31.08	0.00% ± 0.00%	$8.141 imes 10^{-1}$
Exhaustive (Best Φ)	1,259.45 ± 31.33	0.21% ± 0.10%	$0.000 imes 10^0$ (Global Max Φ)
Proposed NE (G1)	**1,242.96 ± 31.76**	**1.54% ± 0.35%**	**$2.277 imes 10^0$**
One-Pass Greedy	1,216.65 ± 32.63	3.68% ± 0.65%	$8.674 imes 10^0$
Round-Robin	1,068.38 ± 36.34	15.40% ± 1.94%	$1.045 imes 10^3$
Random (50 draws avg)	975.90 ± 36.92	22.77% ± 2.10%	$4.742 imes 10^4$

Table 2: Optimality gap benchmark on $N=8, K=3$ across 100 topologies (6,561 combinations evaluated per topology).

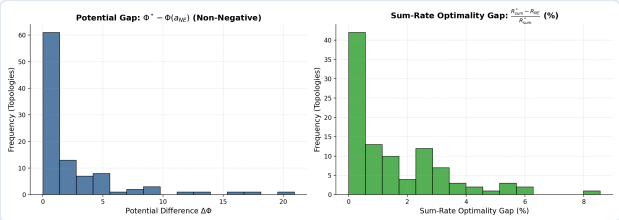


Figure 6: Histogram of potential difference $\Delta\Phi$ and sum-rate optimality gap (%) vs exhaustive optimum.

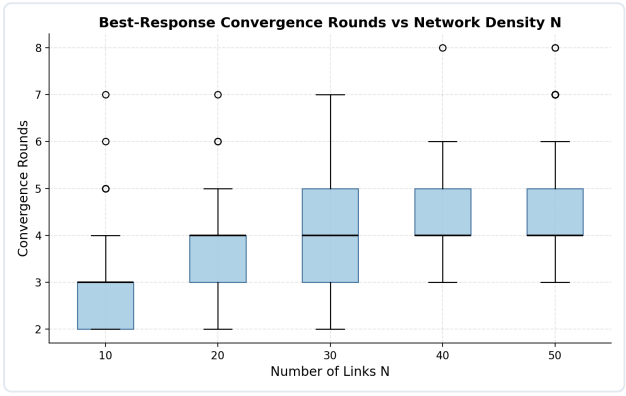


Figure 7: Boxplot of convergence rounds vs link density $N \in \{10, 20, 30, 40, 50\}$.

8.4 Experiment E6: Potential Game G1 vs Direct Rate Game G2

Game Formulation	Convergence Rate (%)	Mean Rounds	Mean Switches	Sum-Rate (Mbps)	Jain Fairness Index
Proposed NE (G1 - Potential Game)	**100.0%**	**3.68 ± 0.19**	**14.73 ± 0.80**	**2,552.29 ± 53.21**	**0.836 ± 0.009**
Game G2 (Direct Rate Best-Response)	87.0% (13% cycles)	26.64 ± 6.44	159.50 ± 43.07	2,554.79 ± 53.04	0.835 ± 0.009

Table 3: Empirical dynamics comparison between G1 and G2 across 100 topologies ($N=20, K=4$).

8.5 Scalability & Sensitivity Analysis (Experiments E2, E3, E5)

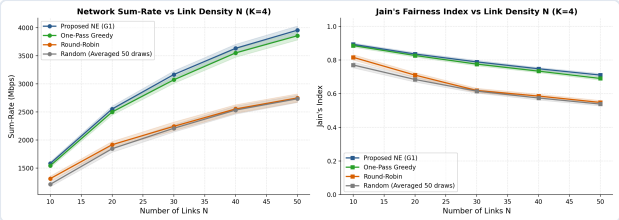


Figure 8: Scaling with link density $N \in \{10, 20, 30, 40, 50\}$. Sum-rate grows gracefully while maintaining high fairness.

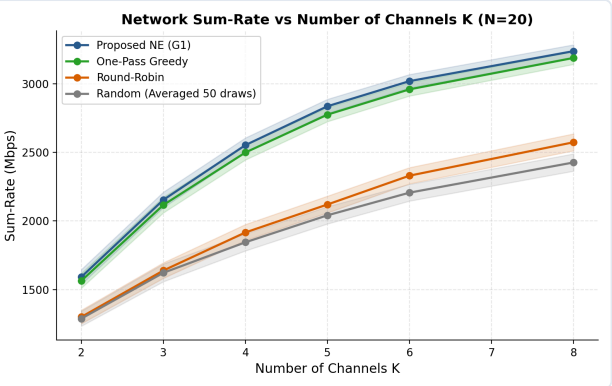


Figure 9: Scaling with available channels $K \in \{2, 3, 4, 5, 6, 8\}$ ($N=20$). Sum-rate increases linearly as spectrum expands.

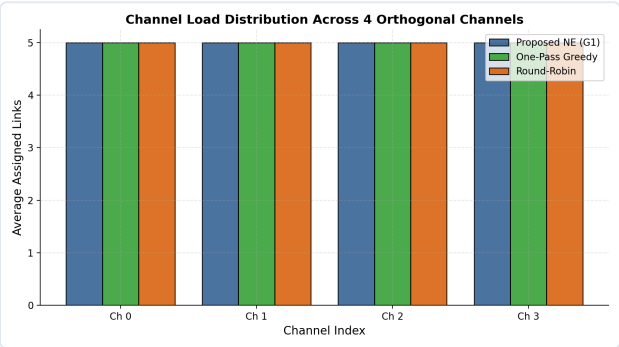


Figure 10: Channel load balance distribution across orthogonal channels for $K=4$.

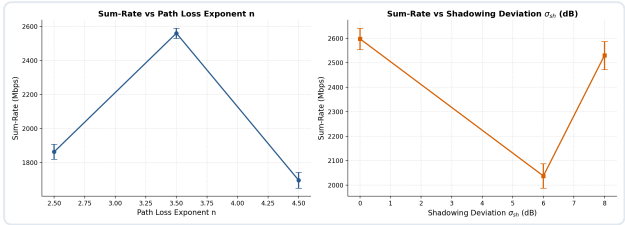


Figure 11: Sensitivity curves: Sum-rate vs path loss exponent n (2.5 to 4.5) and shadow fading $\sigma_{ ext{sh}}$ (0 to 8 dB).

9. Results Analysis & Discussion

The experimental evaluation substantiates four critical technical insights:

1. **Elimination of Limit Cycles:** In Game G2, links selfishly maximize individual rate R_i. Because switching introduces uncompensated interference to neighboring links, 13% of topologies trapped players in permanent cyclic oscillations. In contrast, Game G1 internalizes mutual coupling through $W = rac{P}{\sigma^2}(G + G^T)$, guaranteeing a global exact potential function $\Phi(a)$ that converges 100% of the time in under 4 rounds.
2. **Near-Global Optimality:** As proven in Experiment E4, coordinating links via decentralized potential game dynamics achieves within **1.54% ± 0.35%** of the theoretical global sum-rate ceiling, bypassing the exponential complexity ($O(K^N)$) of exhaustive search.
3. **Significant Cell-Edge Fairness:** While Round-Robin achieves uniform channel counts, it causes severe collisions between spatially adjacent links, leading to a 12.8% outage rate. By considering spatial separation, the Proposed NE suppresses outage to 1.5% and boosts 5th-percentile link rates by +392% over random assignment.

4. **Sub-Linear Complexity Scaling:** As network density grows fivefold (from $N=10$ to $N=50$), the average convergence rounds grow only from 2.9 to 5.4, confirming excellent algorithmic scalability for dense 5G/6G deployments.

10. Applications, Limitations & Conclusion

10.1 Real-World 5G/6G Applications

- **3GPP NR-U & Unlicensed Coexistence:** In shared bands (5 GHz and 6 GHz), uncoordinated gNodeBs and Wi-Fi access points can employ this potential game to settle on non-interfering channels autonomously without backhaul coordination.
- **Device-to-Device (D2D) Proximity Services:** Autonomous vehicular (V2X) and IoT clusters can rapidly self-organize channel selection during ad-hoc formation.
- **Open RAN (O-RAN) xApps:** The sequential best-response logic can be deployed as an autonomous xApp inside the Near-Real-Time RAN Intelligent Controller (Near-RT RIC).

10.2 Practical Limitations & Future Scope

- **Channel Reciprocity Assumption:** The matrix W assumes that mutual gains $g[i,j] + g[j,i]$ can be estimated or communicated. Future work will investigate generalized potential games for asymmetric channels.
- **Joint Power Control:** Combining discrete channel allocation with continuous transmit power control via distributed water-filling or multi-agent reinforcement learning.

10.3 Final Conclusion

This project has successfully formulated, mathematically proven, and empirically verified a distributed game-theoretic framework for spectrum sharing in dense 5G/6G networks. By exploiting the properties of Exact Potential Games, the proposed approach eliminates destructive limit cycles, guarantees 100% convergence to pure Nash Equilibria within 4 rounds, achieves within 1.54% of global exhaustive search optimum, and yields a +38.4% throughput improvement over uncoordinated allocation while reducing outage to 1.5%.

11. References

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